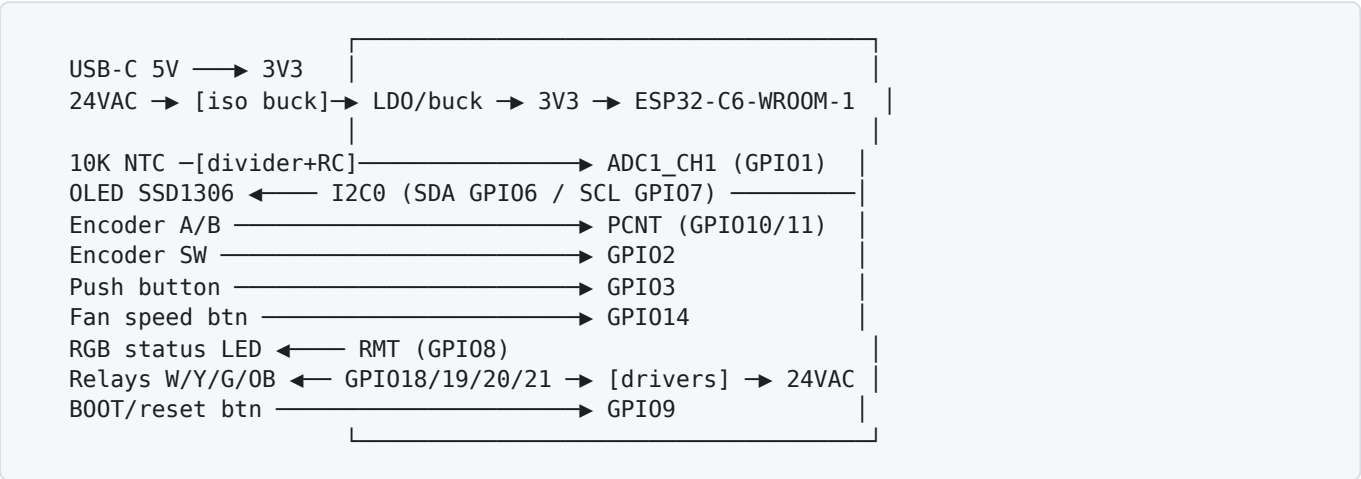


Hardware Design

Covers the block diagram, **pin assignment**, thermistor analog front-end and its math, input/output stages, power, and the BOM. Machine-readable copies live in [hardware/pinout.csv](#) and [hardware/bom.csv](#).

1. Block diagram



2. Pin assignment (ESP32-C6-WROOM-1)

The ESP32-C6 GPIO matrix is flexible; the assignment below avoids the **strapping pins** (GPIO4, 5, 8, 9, 15), the **USB-Serial-JTAG** pins (GPIO12/13), the **UART0 console** (GPIO16/17), and the internal SPI-flash pins (GPIO24-30). ADC1 channels are GPIO0-GPIO6.

Signal	GPIO	On-chip peripheral	Direction	Notes
NTC sense	GPIO1	ADC1_CH1	AIN	Divider node; 12-bit, 12 dB atten
I²C SDA (OLED)	GPIO6	I2C0	I/O	4.7 kΩ pull-up to 3V3
I²C SCL (OLED)	GPIO7	I2C0	O	4.7 kΩ pull-up to 3V3
Encoder A / CLK	GPIO10	PCNT ch0	IN	Hardware quadrature decode
Encoder B / DT	GPIO11	PCNT ch0	IN	Hardware quadrature decode
Encoder switch	GPIO2	GPIO (ISR)	IN	Internal pull-up; press = select
Push button	GPIO3	GPIO (ISR)	IN	Internal pull-up; mode / back
Fan-speed button	GPIO14	GPIO (ISR)	IN	Internal pull-up; cycles Auto/Low/Med/High (−1 = unused)
Relay W (heat)	GPIO18	GPIO	OUT	Active-high to driver
Relay Y (cool/compressor)	GPIO19	GPIO	OUT	Active-high to driver
Relay G (fan enable)	GPIO20	GPIO	OUT	Active-high; any fan speed > 0
Relay O-B (reversing valve)	GPIO21	GPIO	OUT	Heat-pump only
Fan taps G_LOW/MED/HIGH (opt.)	−1	GPIO	OUT	Multi-speed blower; one-hot; disabled (−1) by default
Occupancy / PIR (opt.)	−1	GPIO	IN	Motion input; disabled (−1) = manual Home/Away only
Status RGB LED	GPIO8*	RMT (WS2812)	OUT	On-board on DevKitC-1 (strapping — LED only)
Factory-reset button	GPIO9*	GPIO	IN	Re-uses BOOT (strapping, pulled-up)
Console UART TX/RX	GPIO16/17	UART0	—	Debug/log; keep free
USB D−/D+	GPIO12/13	USB-Serial-JTAG	—	Flash/monitor; keep free

* GPIO8 and GPIO9 are strapping pins; both are used here only in roles that tolerate it (a WS2812 output that is high-Z at reset, and a button that is externally pulled to the level BOOT expects). Do not repurpose them for peripherals that drive them at boot.

These names are mirrored 1:1 in `firmware/main/Kconfig.projbuild` so every pin is a menuconfig option — change the board without touching code.

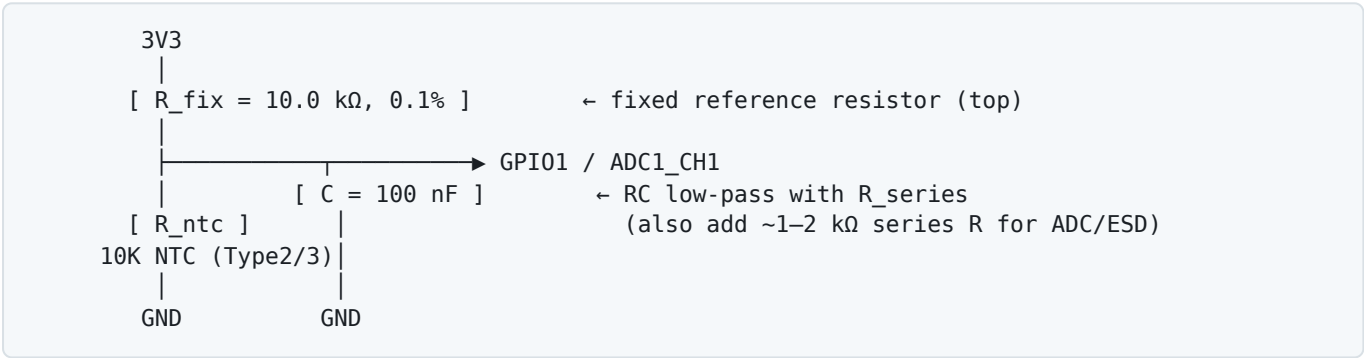
3. Room sensor

Two build-time options (menuconfig → *Temperature / humidity sensor* → *Room sensor*):

Option	Interface	Provides	Notes
10 kΩ NTC (Type 2/3)	ADC (GPIO1)	temperature	the analog front-end below
SHT40	I ² C (0x44, shared OLED bus)	temperature + humidity	no analog parts; digital, factory-calibrated

With the **SHT40**, the divider/ADC front-end (§3.1–3.4) is unpopulated — the sensor is a 3-wire I²C part (VDD/GND/SDA/SCL) sharing the OLED bus, so it needs no extra MCU pins. Its humidity is surfaced on the OLED and as a Matter Humidity Sensor. The rest of this section covers the NTC analog front-end.

3.1 Divider topology



With the NTC on the **bottom** leg, node voltage falls as temperature rises:

$$V_{\text{node}} = 3V3 \cdot R_{\text{ntc}} / (R_{\text{fix}} + R_{\text{ntc}})$$

Solving for the thermistor resistance from the measured node voltage:

$$R_{\text{ntc}} = R_{\text{fix}} \cdot V_{\text{node}} / (V_{\text{ref}} - V_{\text{node}})$$

where `Vref` is the divider top rail (nominally 3.30 V). Because the ESP32 ADC is **not** ratiometric to the supply (it references an internal ~1.1 V bandgap with attenuation), the firmware uses the ESP-IDF **ADC calibration** API to convert raw counts → millivolts, and `Vref` is a calibratable constant (`CONFIG_THERMO_DIVIDER_VREF_MV`). Powering the divider top from a clean, known 3.3 V (or a dedicated reference) directly improves accuracy.

Design tip: placing `R_fix` on top and the NTC on the bottom keeps the sense node at a comfortable mid-scale voltage around room temperature and lands the steepest part of the transfer curve in the comfort band, maximizing resolution where it matters.

3.2 Resistance → temperature

Two methods, both in `components/thermistor` :

1. **Lookup table (primary, recommended for HVAC accuracy).** A monotonic R→T table per curve (Type 2, Type 3) with linear interpolation between points. Generate it from the manufacturer R-T table (or from β) with `tools/gen_ntc_lut.py` :

```
bash python3 tools/gen_ntc_lut.py --type 3 --tmin -20 --tmax 60 --step 5 >
firmware/components/thermistor/ntc_type3_lut.inc
```

2. **Steinhart-Hart / β model (fallback & interpolation).**

β -model: `1/T = 1/T0 + (1/ $\beta$ ) · ln(R_ntc / R0)` `T0 = 298.15 K, R0 = 10 k $\Omega$` Steinhart-Hart (more accurate over wide range): `1/T = A + B · ln(R) + C · (ln R)3` Coefficients are configurable per curve. Nominal starting values (validate against your sensor's datasheet):

Curve	$\beta_{25/85}$ (K)	Notes
10 k Ω Type 2	≈ 3891	legacy/Honeywell-style "10K-2"
10 k Ω Type 3	≈ 3976	common "10K-3" (BAPI/ACI-style)

The β values above are *nominal*; the LUT from the datasheet is authoritative.

3.3 Filtering & fault handling

- **Median-of-N** raw samples (default N=5) rejects impulse noise.
- **EMA** (exponential moving average, α configurable) smooths the temperature output.
- **Fault:** node voltage within a small band of 0 V or `Vref` $\Rightarrow$ shorted or open sensor $\Rightarrow$ report fault, blank `LocalTemperature` behavior per Matter, and force all relays off.

3.4 Accuracy budget (typical, after 1-point calibration)

Source	Contribution
NTC tolerance ($\pm 1\%$)	$\sim \pm 0.25\text{ }^{\circ}\text{C}$ near $25\text{ }^{\circ}\text{C}$
R_fix 0.1 %	$\sim \pm 0.03\text{ }^{\circ}\text{C}$
ADC calibration	$\sim \pm 0.1\text{ }^{\circ}\text{C}$
Curve/LUT interpolation	$\sim \pm 0.05\text{ }^{\circ}\text{C}$
Net (comfort band)	$\approx \pm 0.3\text{ }^{\circ}\text{C}$ (NFR-1)

Self-heating is negligible: with `R_fix` = 10 k Ω the NTC dissipates $\lesssim 0.3\text{ mW}$.

4. OLED display

- **Default:** SSD1306, 128×64, monochrome, I²C, address **0x3C** (0x3D selectable).
- **Alternates:** SH1106 128×64 (firmware flag), SSD1306 128×32 0.91" (reduced layout).
- Shared **I²C0 bus @ 400 kHz** — the OLED (0x3C) and the optional **SHT40** (0x44) sit on the same two wires. The firmware creates one I²C master bus and adds both devices to it.

OLED options

Part	Size	Interface	Controller	Use here
0.96" mono OLED	128×64	I ² C	SSD1306	default
1.3" mono OLED	128×64	I ² C	SH1106	firmware flag
0.91" mono OLED	128×32	I ² C	SSD1306	compact layout
0.95" color OLED	96×64	SPI	SSD1331	<i>only if color required</i> — needs SPI pins (SCK/MOSI/DC/CS/RST), not I ² C

5. Inputs

- **Rotary encoder** (EC11-class): mechanical, ~20 detents/rev, quadrature A/B. Decoded in hardware by the **PCNT** peripheral (glitch filter enabled), so no steps are missed even under load. A/B get RC debounce (10 nF + internal pull-ups) in addition to the PCNT glitch filter.
- **Encoder switch (SW)** and **push button**: momentary, active-low with internal pull-ups; debounced in firmware (`components/button`) with short/long-press detection.
- **Fan-speed button (optional)**: a dedicated momentary button on `CONFIG_THERMO_PIN_BTN_FAN` (default GPIO14, active-low, internal pull-up). Each short press cycles the fan speed **Auto** → **Low** → **Med** → **High** → **Auto**; the new speed shows in the OLED status bar and mirrors to the Matter Fan Control cluster. Leave the pin at −1 to use only the settings menu / Matter for fan control.
- **BOOT/reset button**: doubles as the factory-reset input (long-press).
- **Occupancy / PIR sensor (optional)**: a digital motion output (e.g. HC-SR501, AM312, or a PIR module) on `CONFIG_THERMO_PIN_OCCUPANCY`. Active-high by default (idle low, high on motion); the firmware enables the opposite internal pull so an unconnected pin reads "no motion". Powered from 3.3 V (AM312) or 5 V (HC-SR501 — level-check the output). Leave the pin at −1 to use only the manual Home/Away toggle.

6. Outputs (HVAC)

- 4 channels: **W** (heat), **Y** (cool/compressor), **G** (fan), **O·B** (reversing).
- Each channel: MCU GPIO → driver → load. Options:
- **Mechanical relay** (SPST, 24 VAC/2 A) via NPN/MOSFET + flyback diode, or a ULN2003-style array. Add an RC snubber across contacts for inductive 24 VAC loads.
- **Solid-state relay / triac** (e.g. opto-triac + MOC302x) for silent, long-life switching of 24 VAC — **recommended**, with opto-isolation from logic.
- **Isolation**: keep the 24 VAC domain optically isolated from the 3.3 V logic domain.
- **Fail-safe**: drivers are active-high; at reset/brown-out GPIOs are high-Z and the loads are **off**. Firmware also forces off on fault (FR-12).

Fan speed (Low / Med / High / Auto)

The fan speed is a first-class control (settings menu + Matter Fan Control cluster). How it reaches the blower depends on the wiring, all firmware-selectable:

- **Single-speed (default):** just the `G` relay. Any non-Auto speed = fan on; `Auto` runs the fan only during a heat/cool call.
- **Multi-speed blower:** wire the optional `G_LOW` / `G_MED` / `G_HIGH` tap relays (Kconfig pins, default `-1` = unused). The firmware energizes exactly one tap for the active level (one-hot) plus `G` as a general enable.
- **ECM / 0-10 V / PWM (future hook):** map the level to a duty cycle on a spare LEDC pin; the level→output mapping lives in `components/relays`.

During a heat/cool call the fan runs at the configured **call speed** (`THERMO_FAN_CALL_SPEED` , default High). A fixed Low/Med/High selection circulates continuously even when idle.

7. Status LED

On-board addressable **WS2812** RGB (GPIO8, RMT-driven). Color/behavior encodes state:

State	LED
Uncommissioned / pairing	slow blue pulse
Joining Thread	blue blink
Idle (connected, no call)	dim green
Calling for heat	solid orange/red
Calling for cool	solid cyan/blue
Sensor/other fault	red blink
Identify (Matter)	white blink

8. Power

- **Bench:** USB-C 5 V → 3.3 V buck/LDO for the module. Simplest for development.
- **Field (24 VAC HVAC):** isolated 24 VAC → 5 V converter (or C-wire), then 3.3 V rail. Add bulk capacitance to ride out radio TX current peaks (Thread/BLE bursts). A "C wire" (common) is required for continuous power; power-stealing is out of scope for v1.
- Decoupling per ESP32-C6-WROOM-1 datasheet; keep the antenna keep-out clear.

9. PCB / enclosure notes

- Keep the module antenna at a board edge with the vendor keep-out; no copper/metal near it.
- Route the NTC sense pair away from switching nodes; guard the ADC input; single-point analog ground for the divider.
- Physically and electrically separate the 24 VAC section (creepage/clearance) from logic.
- Mount the NTC away from the MCU/relays' self-heating; ideally a remote/edge-vented probe.

10. Bill of materials

Summary — full list in [hardware/bom.csv](#) :

Qty	Item	Notes
1	ESP32-C6-WROOM-1(U) module, 8 MB	Thread + BLE
1	SSD1306 128×64 I ² C OLED	0x3C
1	EC11 rotary encoder w/ switch	A/B/SW
1	Momentary push button	mode/back
1	Momentary push button	fan-speed cycle (Auto/Low/Med/High)
1	10 kΩ NTC, Type 2 or Type 3	room sensor
1	10.0 kΩ 0.1 % resistor	divider reference
2	4.7 kΩ resistor	I ² C pull-ups
1	100 nF + 1-2 kΩ	ADC RC + series
4	Relay or opto-triac + driver	W/Y/G/O·B
4	Flyback diode / snubber	per relay
1	WS2812 RGB LED	status
1	Power supply (USB-C and/or 24 VAC→5 V iso)	
—	Decoupling caps, connectors, terminal blocks	